v1.8.1 — Document identity, fillet verification, and optimization

Five releases' worth of work since **v1.4.1**, driven by two long real-world modelling sessions. **60 tools** now, up from 50.

The theme across all of it: *"no error" is not "it worked."* Most of the expensive failures in this project weren't crashes — they were operations that reported success while doing nothing, doing it to the wrong document, or silently destroying geometry elsewhere. Most of this release is about making those visible.

⚠️ Upgrading — read this

Two files, two different folders. They are separate installs and updating one does not update the other:

File	Goes to
`fusion_mcp_server.py`	your project folder (whatever `claude_desktop_config.json` points at) → restart Claude Desktop
`MCPListener.py`	`%appdata%\Autodesk\Autodesk Fusion 360\API\AddIns\MCPListener\` → Shift+S → MCPListener → **Stop → Run**

Then run `ping_fusion`. It now reports **both** versions and warns if they differ — expect `server v1.8.1 | listener v1.8.1`. A stale add-in reports `PROTOCOL MISMATCH` instead of silently returning empty payloads.

Highlights

Document identity (1.6.0) — the most damaging bug class, fixed

Claude couldn't reliably tell which document was open. Symptoms: overwriting an unrelated design, starting a new document instead of continuing in the open one, and reporting saves that never happened. Four compounding causes:

- **`doc.dataFile` routes through `app.data`**, which can throw for an entire session on some builds. A *saved* document therefore looked unsaved, and the old logic responded by calling `saveAs` — **creating a duplicate**. `save_document` now tries the non-destructive `doc.save()` first when lineage is unreadable, and refuses to fall back to `saveAs` without explicit `force_new_lineage=True`.

- **Identity was a stale cache** that only refreshed after a job ran. Switching documents in the Fusion UI left it reporting the previous one. It now refreshes on demand and reports its own snapshot age and a `fresh` flag.
- `isSaved` **was the sole branch signal.** Every signal is now read independently so contradictions are visible.
- **No visibility of open documents.** New `document_report()` — an authoritative live read listing every open document, distinguishing *"never saved"* from *"cannot tell"*. New `activate_document()` binds work to the right file.

Saves now print **CONFIRMED** only when an independent signal agrees (version bump / modified cleared / lineage appeared), otherwise **UNCONFIRMED**.

Verification everywhere (1.6.1)

Audited for the same pattern elsewhere and found four more:

- **40 tools assumed** `app.activeProduct` **is a Design.** It isn't in the Drawing/CAM/Render workspace. All now route through a guard that names the actual product type instead of failing obscurely.
- **Exports announced success without checking.** `em.execute()` 's return value was being *discarded*. STEP/STL/DXF now verify the return value **and** that the file exists with non-zero size.
- `set_parameter` **never read back.** Fusion silently clamps expressions it dislikes, so a parameter could report as 50 mm while holding 2 mm. Now reads back and reports `MISMATCH` .
- **Joints reported success without verifying.** Now checks the joint count increased.
- `clear_document` — the most destructive tool — checked nothing. It now prints the target document name and lineage before deleting.

Fillets that can actually be aimed (1.5.0)

`which=all/vertical/top` couldn't express what real fillets need. New `select_edges(face_pair="Cylinder|Cone", z_min_mm=…, min_len_mm=…)` returns templds straight into `fillet_edges_smart` . `list_edges` now reports each edge's adjacent surface-type pair.

`fillet_edges_smart` now verifies **topology as well as volume** — two opposite failure modes:

- a blend that propagates and **adds** geometry far away (volume overshoot);
- an over-large radius that **consumes** a narrow feature (a 1.3 mm rim annulus under a 2 mm fillet) — where the volume delta looks perfectly reasonable while the feature is rounded away.

New `fillet_preview()` flags at-risk faces before committing. A **zero volume delta is now reported as a result, not a failure** — it means the surfaces already meet tangentially.

Escape hatches (1.5.0)

- `force_delete_feature()` — resolves a mutual-reference deadlock where two features each blame the other and neither suppress nor delete works in any order. Rolls the marker back before deleting, which is what the Fusion UI does.

- `sweep_tube_check()` — `ASM_SELF_INTER` blames the profile, but on a circular profile the *path* is the cause. Measures minimum path curvature against tube radius.
- `respline()` — rebuild a fitted spline with new fit points (Fusion can move but not add them). Honest about breaking downstream references.

Performance (1.7.0 / 1.8.0)

Measured, not guessed:

Change	Effect
Lazy identity refresh	Removed a 4-second main-thread heartbeat (~21,600 interrupts/day). **Idle host cost now zero.**
`run_batch()`	Fuses N operations into **one round trip and one Fusion job**
Docstring trim + `fusion_help()`	**~811 tokens saved on every request** (~40k per 50-turn session)
`view_model` **resolution**	Default 1200×900 → 800×600: **~800 tokens saved per screenshot**
Terse-on-success output	Detail hidden on clean success; **always shown in full on any warning**
`cross_section` **bisection**	Boundary precision **1.94 mm → 0.001 mm** for 16 extra kernel queries
Face-iteration caps	Bounded at 4,000 faces instead of stalling on dense models

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Fixes in 1.8.1

Two regressions the optimization work itself introduced, found by stress-testing rather than assuming:

- `run_batch` **could fire a destructive operation during collection.** It used a *denylist*, so anything unlisted executed for real during what should have been a dry pass — batching `clear_document` actually wiped the document, then reported the batch as failed. Now an explicit **allowlist** of 27 geometry/parameter tools, with all operations validated *before* collection begins, the async path hard-refused while collecting, and a lock so concurrent calls can't be swallowed.
- **Terse mode hid the body of report tools.** `document_report`'s entire output is indented, so the tool built to prevent wrong-document overwrites was reduced to one line. Compaction is now opt-in per call; 18 report tools opt out.

Other fixes since 1.4.1

- Degenerate edges (zero-length, <2 faces) no longer crash edge iteration with `Bad index parameter` — guards are inside the listener helpers.
- `explain_fillet_error` gained nine more ASM codes (`ASM_ABL_NO_GEOM_EF` , `ASM_ABL_CVXTY_VIOLATION` , `ASM_BL_NO_VTX_GEOM` , `EDGE_REFERENCE_LOST` , …).
- `refresh_geometry()` — `computeAll()` doesn't guarantee a fresh `body.volume` ; identical readings after a sketch edit can make a correct edit look failed.
- `path_report()` — a mis-chained path silently produces a short sweep; segment count is the only cheap tell.
- Tool docstrings carry a version stamp, because Claude Desktop caches tool schemas at startup and a corrected docstring can still *display* stale.

Known limitations

- **`open_document` / `list_recent_designs` / `list_versions` require `app.data` .** When it's broken for a session there is no workaround — open the design in the Fusion UI.
- **`defineAsFreeMove` and `defineAsOffset` are refused** by the listener; the former hard-crashes the Fusion host.
- **This is verified against mocks, not a live Fusion session.** The `adsk` API surface is exercised through simulated objects, so treat the first run of any release as a shakedown and keep your work saved.